Petra Michael

Software Developer & Researcher

Jakarta, Indonesia

petramichael.vercel.app

linkedin.com/in/aimatochysia

github.com/aimatochysia

SUMMARY

Passionate software developer and researcher from Indonesia with experience in building cross-platform applications and secure, data-driven solutions. Having experience in full-stack development using Flutter, Python, and JavaScript/React, as well as embedded and smart card systems leveraging APDU communication and JavaCard technologies. Able to implement post-quantum cryptography and adhering to security and compliance standards under NDA. Committed to delivering high-performance software with robust security practices, following industry and proprietary standards in confidential environments.

EDUCATION

Bina Nusantara University (BINUS) Aug 2022- Aug 2026

Bachelor of Computer Science, School of Computer Science

Bandung, West Java, Indonesia

WORK EXPERIENCE

Software Development Engineering Intern Feb 2025 - Feb 2026

IDEMIA - South Jakarta, Indonesia (On-site)

- Handled internal software development testing for identity and security solutions, including FIDO and other payment/security standards under NDA
- Developed and maintained automated testing frameworks using Java, Jython, and Jenkins to improve reliability and efficiency
- Worked with APDU protocols, EMV standards, and JavaCard technologies for smart card interfaces in secure environments
- Collaborated with development teams to enhance software implementation and testing strategies, ensuring high performance and compliance with industry standards

Research Assistant Jun 2023 - May 2024

Bina Nusantara University - Bandung, West Java, Indonesia (Hybrid)

- Dataset cleaning for AI training application in fishpond water quality prediction based on image input
- Applied machine learning techniques and Python for data processing and model development
- Contributed to research initiatives combining computer vision with environmental monitoring

Website Maintainer May 2023 - Apr 2024

Bina Nusantara University - Bandung, West Java, Indonesia (Hybrid)

- Managed and maintained apski.net, a collaboration between universities with School of Business programmes across Indonesia
- Project is tied and supported by the Ministry of Cooperation and Small Businesses of Indonesia (KemenkopUKM)
- Added and fixed web content, ensuring optimal user experience and site functionality

ORGANIZATIONAL EXPERIENCE

Treasurer
Feb 2024 - Jan 2025

Nippon Club - BINUS University - Bandung, West Java, Indonesia (Hybrid)

- Managed financial operations for future event planning and organizational expansion
- Oversaw budgeting, financial accounting, and treasury accounting for non-profit operations

Activist
Feb 2023 - Jan 2024

Nippon Club - BINUS University @Bandung, West Java, Indonesia

- Led a charity program and oversaw the general expo of the organization
- Contributed to event management and social change initiatives

Academic Events Manager
Feb 2024 - Jan 2025

HIMTI BINUS University - Bandung, West Java, Indonesia (Remote)

- Led a team of education activists to teach programming hard skills
- Organized training sessions for School of Computer Science members and the public
- Managed operational control and operations for academic events

Activist
Feb 2023 - Jan 2024

HIMTI BINUS University - Bandung, West Java, Indonesia (Hybrid)

- Taught programming hard skills to School of Computer Science members and the public
- Contributed to event management and project management initiatives

TECHNICAL SKILLS

Programming Languages: Python, Dart, JavaScript, TypeScript, Java, Jython, C, C++, Rust, GDScript

Frameworks & Libraries: Flutter, React.js, Next.js, Node.js, Flask, Godot Engine, Chrome Extension APIs

Data & Analysis: Jupyter Notebook, Pandas, NumPy, yfinance, YOLO, CNN

DevOps & Tools: Git, GitHub Actions, Vercel, CI/CD Pipelines

SOFT SKILLS

Programming Languages: Python, Dart, JavaScript, TypeScript, Java, Jython, C, C++, Rust, GDScript

Frameworks & Libraries: Flutter, React.js, Next.js, Node.js, Flask, Godot Engine, Chrome Extension APIs

Data & Analysis: Jupyter Notebook, Pandas, NumPy, yfinance, YOLO, CNN

DevOps & Tools: Git, GitHub Actions, Vercel, CI/CD Pipelines

PERSONAL PROJECTS

Q-Safe Vault
github.com/aimatochysia/qsafevault

Dart, Flutter, Rust FFI | Post-Quantum Cryptography

Secure, local-first password manager featuring post-quantum cryptography. Implements NIST-standardized Kyber ML-KEM 768 for key encapsulation and Dilithium3 for digital signatures. Features AES-256-GCM encryption, Argon2id key derivation, zero-trust sync with no central server, and hardware security module support (TPM2.0, SoftHSM, Secure Enclave).

FicBatch
github.com/aimatochysia/FicBatch

Dart, Flutter | Cross-Platform Mobile & Desktop

Cross-platform Archive of Our Own (AO3) browser and reading app. Features embedded web browsing with dark mode, dedicated reader with autosave, local library management, background sync for chapter updates, and reading history tracking.

Stock Screener (JKSE)
github.com/aimatochysia/stock-screener

Python, React.js, GitHub Actions | Data Pipeline & Web

Automated stock screening system for Jakarta Stock Exchange (~900 stocks). Implements distributed data pipeline with daily OHLCV collection, technical analysis (RSI, ATR, SMA, support/resistance), and interactive React frontend. Features automated GitHub Actions workflows for data processing across multiple repositories.

Wallpaper Engine Code Clock
github.com/aimatochysia/Wallpaper-Engine-Code-Clock

HTML, JavaScript, CSS | Desktop Customization

Wallpaper Engine wallpaper displaying a live clock styled as programming language syntax. Supports multiple languages including Python, C, C++, C#, Java, and JavaScript. Published on Steam Workshop. As of currently have 25K+ subscribers on Steam.

INTERNATIONAL PUBLICATIONS

Data Article: Aerial images and water quality dataset for fishpond's condition monitoring (DOI:10.1016/j.dib.2023.110009)

Conference Paper: Investigation on Correlation of Water Quality Data with Aerial Images (DOI:10.1109/ICDXA61007.2024.10470473)

LANGUAGES

Indonesian: Native

English: Professional Proficiency

INTERESTS

Post-Quantum Cryptography, Financial Technology, Embedded Systems, IoT Projects, Game Development, Machine Learning, Material Science, 3D Animation & Illustration (Blender3D, Pixel Art)